

1. Table Creation (CREATE): Write the SQL statements to create a database named “employee” and the following tables based on the provided schema:

☉ Departments ☉ Location ☉ Employees

```

  >Run
✓ 4  CREATE DATABASE employee; 4ms AffectedRows: 1
  >Run
✓ 5  use employee; 3ms
  >Run | ⌵Select
✓ 6  CREATE table departments(
  7  ✦ department_id INT PRIMARY KEY,
  8  department_name VARCHAR(100)
  9  ); 15ms
  >Run | ⌵Select
✓ 10 CREATE TABLE location (
  11 location_id INT PRIMARY KEY,
  12 location VARCHAR(30)
  13 ); 45ms
  >Run | ⌵Select
✓ 15 CREATE TABLE employees (
  16 employee_id INT PRIMARY KEY,
  17 employee_name VARCHAR(50),
  18 gender ENUM('M', 'F'),
  19 age INT,
  20 hire_date DATE,
  21 designation VARCHAR(100),
  22 department_id INT,
  23 location_id INT,
  24 salary decimal(10,2)); 45ms

```

2. Table Alteration (ALTER): Consider the following scenarios and write the SQL statements to alter the structure of the tables accordingly:

- ☉ Add a new column named "email" to the Employees table to store employee email addresses.
- ☉ Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- ☉ Drop the “age” column from the Employees table.
- ☉ Rename the “hire_date” column to “date_of_joining”.

```

  26  alter table employees add column email varchar(100); 44ms
  27  alter table employees modify column designation text; 31ms
  28  alter table employees drop column age; 44ms
  29  alter table employees change column hire_date date_of_joining date; 7ms
  30  rename table departments to Departments_Info; 35ms
  31  Rename table location to Locations; 46ms
  32  truncate table employees; 50ms
  33  drop table employees; 46ms
  34  drop database employee; AffectedRows: 2 53ms

```

3. Database Recreation:

● Drop the 'employee' database if it exists and recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.

```

  35  drop database if exists employee; 25ms AffectedRows: 3
  36  create database employee; 4ms AffectedRows: 1
  37  use employee; 3ms

```

4. Departments Table: ① Ensure that the "department_id" uniquely identifies each department.
- ② Set up constraints on the "department_name" to avoid duplicate and null entries.

```
✓ 38 create table departments (  
39     department_id int primary key,  
40     department_name varchar(100) not null unique  
41 ); 48ms  
42
```

5. Location Table: ① Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially. ② Implement constraints to prevent the insertion of null and duplicate locations.

```
✓ 42 create table location (  
43     location_id int auto_increment primary key,  
44     location varchar(30) not null unique  
45 ); 49ms  
46
```

6. Employees Table: ① Guarantee that each employee has a distinct identifier. Create a restriction to ensure that the employee's name is always provided. Limit the acceptable values. Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

```
✓ 47 create table employees (  
48     employee_id int primary key,  
49     employee_name varchar(50) not null,  
50     gender enum('M', 'F'),  
51     age int check (age >= 18),  
52     hire_date date default (current_date),  
53     designation varchar(100),  
54     department_id int,  
55     location_id int,  
56     salary decimal(10,2),  
57  
58     foreign key (department_id)  
59         references departments(department_id),  
60  
61     foreign key (location_id)  
62         references location(location_id)  
63 ); 53ms  
64
```

Field	Type	Null	Key	Default	Extra
varchar	blob	varchar	varchar	blob	varchar
> employee_id	int(11)	NO	PRI	(NULL)	
> employee_name	varchar(50)	NO		(NULL)	
> gender	enum('M','F')	YES		(NULL)	
> age	int(11)	YES		(NULL)	
> hire_date	date	YES		curdate()	
> designation	varchar(100)	YES		(NULL)	
> department_id	int(11)	YES	MUL	(NULL)	
> location_id	int(11)	YES	MUL	(NULL)	
> salary	decimal(10,2)	YES		(NULL)	

Field	Type	Null	Key	Default	Extra
varchar	blob	varchar	varchar	blob	varchar
> department_id	int(11)	NO	PRI	(NULL)	
> department_name	varchar(100)	NO	UNI	(NULL)	

Problems Output Debug Console Terminal Ports Filter (e.g. text, !excludeText, t...)

```

2026-09-28 12:46:56 [DEBUG] Closing connection 1790577845869_meta
2026-09-28 12:47:36 [ INFO] Executing: DESCRIBE employees
2026-09-28 12:47:36 [ INFO] Result: 9 rows retrieved in 42ms
2026-09-28 12:50:34 [ INFO] Executing: DESCRIBE departments
2026-09-28 12:50:34 [ INFO] Result: 2 rows retrieved in 43ms

```

Run

66 DESCRIBE location; 42ms

67

Field	Type	Null	Key	Default	Extra
varchar	blob	varchar	varchar	blob	varchar
> location_id	int(11)	NO	PRI	(NULL)	auto_increment
> location	varchar(30)	NO	UNI	(NULL)	

Problems Output Debug Console Terminal Ports Filter (e.g. text, !excludeText, t...)

```

2026-09-28 12:47:36 [ INFO] Result: 9 rows retrieved in 42ms
2026-09-28 12:50:34 [ INFO] Executing: DESCRIBE departments
2026-09-28 12:50:34 [ INFO] Result: 2 rows retrieved in 43ms
2026-09-28 12:51:18 [ INFO] Executing: DESCRIBE location
2026-09-28 12:51:18 [ INFO] Result: 2 rows retrieved in 42ms

```